

State-Aware Rollout Engine

multi-turn personalized dialogue synthesis

Persona Profile

"Demographic Information" : {.....}
"Career & Work Identity" : {.....}
"Core Values & Beliefs" : {.....}
"Lifestyle & Daily Routine" : {.....}
"Cultural & Social Context" : {.....}
"Hobbies & Interests" : {.....}

Summary: "A 33-year-old
training manager in Northridge, CA..."

Scenario Constructors

- Lifelong**
identity, development, life planning
- High-frequency**
daily routines, recurring needs
- Affective**
grief, uncertainty, emotional salience

Behavior Controller

info-seeking info-processing procedural
content-creation social meta



User Simulator
Simulates user utterances and updates state



Evolving User State

Context ← what happened
Stable ← who they are
Transient ← how they feel now



Oracle Assistant
Generates assistant response

Dialogue history (Turn t -> Turn $t+1$)



User: I'm overwhelmed.



Assistant: Let's make a plan

Quality Check



schema



Structure



State completeness



Profile binding



persona consistency



Conflict detection



Dialogue trajectories



State Traces



Synthetic QA pairs

Training views



Turn t

User state (trace)



User state

...



Assistant

...



Teacher / Oracle view
sees latent state + dialogue



Student view
sees observable dialogue only



Learns from observable dialogue only

Output

personalized trajectories,
state traces, and distilled supervision

Input

persona profile + scenario seeds